

BAHRIA UNIVERSITY

Karachi Campus — Department of Computer Science

Rental Property Management Information System (MIS)

A Web-Based Platform for Centralized Property and Rental Operations Management

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Rental Property Management Information System: A Web-Based Platform for Centralized Property Operations

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***Abstract*—This paper presents the design and implementation of a web-based Rental Property Management Information System (MIS) built using the Django framework, MySQL, and the AdminLTE 3 interface theme. The system replaces fragmented manual processes — paper ledgers, spreadsheet-based rent tracking, and informal communication channels — with a secure, role-based digital platform. It manages the complete lifecycle of property operations: from property and unit registration to tenant onboarding, lease management, automated monthly invoicing, payment recording, maintenance ticketing, owner settlement generation, and system-wide audit logging. Role-Based Access Control (RBAC) enforces strict data isolation across five user roles: Admin, Manager, Staff, Owner, and Tenant. The outstanding balance for any tenant is computed using the formula $B = (P \times M) - \sum_{i=1}^{n} R_i$, ensuring real-time financial accuracy. The system demonstrates how MIS principles — transaction processing, decision support, and enterprise-level data management — can be applied to solve real-world operational inefficiencies in the property management domain.**

***Index Terms*—Management Information System, Property Management, Role-Based Access Control, Django, MySQL, Rental Management, HTMX, Transaction Processing System**

I. Introduction

Management Information Systems (MIS) are information systems that support decision-making and operational management within organizations by collecting, processing, storing, and communicating data [1]. In the property management domain, the absence of a centralized information system results in recurring operational failures: missed payments, untracked maintenance requests, and lack of accountability across staff roles. This project implements a comprehensive MIS for property and rental businesses aligned with the core MIS concepts covered in CSC-458: Transaction Processing Systems (TPS) for daily operational data entry, Decision Support through real-time dashboards and KPI cards, and Enterprise-level data management through a normalized relational database with enforced integrity constraints [2]. The system is developed as an alternative semester project submission for CSC-458, following the project creation methodology defined in the course: Inception, Requirements Gathering (SRS), System Modeling (ERD/DFD/DB Schema), Implementation (working Django application), and Deployment.

A. Motivation

Property management in Pakistan — particularly in cities like Karachi — is largely conducted via WhatsApp groups, paper receipts, and Excel spreadsheets. This leads to lost payment records, duplicate invoicing, and zero audit trail when disputes arise.

A purpose-built MIS directly addresses these pain points.

B. Objectives

- Design a normalized relational database schema (13 tables, 3 SQL views) covering all operational entities.
- Implement RBAC with five distinct user roles, each with filtered views and permissions.
- Automate monthly invoice generation with duplicate-prevention constraints at the database level.
- Provide real-time financial calculations using the outstanding balance formula.
- Generate downloadable PDF invoices, receipts, and owner settlement reports.
- Maintain a permanent, tamper-evident audit log of all significant system events.

II. Problem Statement

Traditional property management in small-to-medium businesses suffers from several well-documented operational problems:

1) **No audit trail:** Payments recorded verbally or in personal notebooks are untraceable if a dispute occurs.
2) **Duplicate invoicing:** Without a constraint-enforced system, the same month can be billed twice.
3) **Role confusion:** All staff access the same data, violating the principle of least privilege.
4) **Manual reconciliation:** Calculating owner remittance is prone to arithmetic errors.
5) **Reactive maintenance:** There is no structured ticketing system.

These problems align with the business challenges MIS is designed to solve [1].

III. Background and Related Work

Several open-source property management systems were analyzed during requirements gathering:

MicroRealEstate (MRE) [4] employs a microservices architecture. Its phased build philosophy informed our development priority ordering.

Nyumbani (Rental HMS) [5] demonstrates atomic database transactions for tenant admission and emphasizes audit logging.

Tenant Manager [6] provides a practical implementation of the cumulative outstanding balance formula.

IV. System Design and Architecture

A. Technology Stack

Table I summarizes the selected technology stack.

TABLE I: System Technology Stack

Layer	Technology
Backend	Django 5.x (Python)
Database	MySQL 8.x via XAMPP
Frontend	AdminLTE 3 + Bootstrap 4
Interactivity	HTMX + Alpine.js
PDF Gen	WeasyPrint
Data Tables	DataTables.js

B. Database Schema Overview

The database consists of 13 tables organized into 8 application modules. The central relational chain is:

$$\text{Property} \rightarrow \text{Unit} \rightarrow \text{Lease} \rightarrow \text{Invoice} \rightarrow \text{Payment}$$

Table II lists all entities.

TABLE II: Database Entities and Primary Constraints

Entity	App	Key Constraint
Location	properties	`UNIQUE(name)`
Property	properties	FK to Location, Owner
Unit	properties	`UNIQUE(property, unit_no)`
Tenant	tenants	`UNIQUE(cnic)`, soft-delete
Lease	leases	Trigger: 1 active per unit
Invoice	finance	`UNIQUE(lease, period_month)`
Payment	finance	`CHECK(amount > 0)`
MonthlySettlement	finance	FK to OwnerContract
OwnerContract	contracts	Commission rate, time-bound
ContractProp	contracts	M2M: Contract $\leftrightarrow$ Property
MaintTicket	maintenance	Trigger: CRITICAL status
AuditLog	audit	SET NULL on actor delete
SysSettings	dashboard	Single-row config table

C. Key Database Constraints

Two constraints deserve particular attention: **Invoice uniqueness:** `UNIQUE(lease, period_month)` prevents generating two invoices for the same month. **Active leases:** A trigger ensures only one active lease exists per unit.

D. Data Flow — Monthly Rent Cycle

The core operational workflow follows a TPS pattern. Figure 1 illustrates the Level-1 Data Flow Diagram.

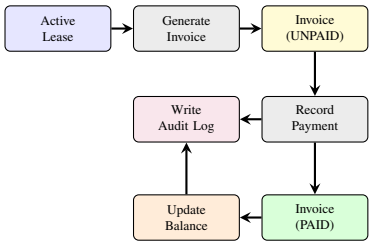


Fig. 1: Level-1 DFD: Monthly Rent Collection Process

V. Financial Model and Key Equation

A. Outstanding Balance Formula

The system computes each tenant's outstanding balance dynamically:

$$B = (P \times M) - \sum_{i=1}^{n} R_i \tag{1}$$

where B = balance, P = rent, M = months elapsed, and R_i = payments.

B. Owner Net Remittance Equation

$$N = G - \left(G \times \frac{C}{100}\right) - K \tag{2}$$

where N = net remittance, G = gross rent, C = commission, and K = maintenance costs.

VI. Results and System Output

A. Dashboard Key Performance Indicators

The Admin dashboard displays live KPI cards.

TABLE III: Sample KPI Card Output (Test Dataset)

Metric	Value
Total Properties	6
Total Units	38
Occupied Units	29 (76.3%)
Monthly Revenue	PKR 8,45,000

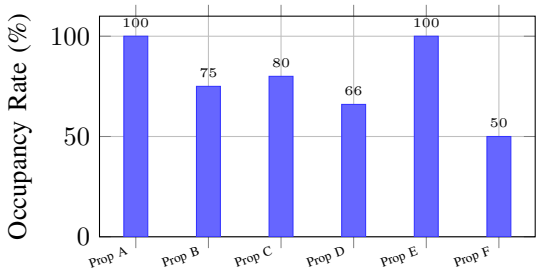


Fig. 2: Occupancy Rate per Property (Test Dataset)

B. Occupancy Rate by Property

VII. Conclusion

This paper presented a Rental Property MIS. Key contributions include a normalized schema, RBAC, and immutable audit logs. The system demonstrates how MIS principles produce reliable operational systems.

References

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